JEFFREY LIM

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Objective

Graduating biomedical/computer engineering student with experience developing machine learning pipelines and medical device work experience at Stryker. Currently developing my thesis project implementing a machine learning pipeline to predict hand-arm vibration risk for power tool operators. Seeking an engineering role incorporating my engineering, medical devices, or software development background. Willing to relocate for a role and have valid Ontario's driver's license.

Skills Summary

- Work experience at Stryker providing **quality control, calibration & repair** for electromechanical medical devices
- **2+ years** developing **Python** machine learning pipelines to identify & predict injury risk for power tool users
- Applied **HTML & Javascript** to develop front-end features in professional (Stryker, Tootyr), and personal projects
- Developed **computer/sensor-**based testing apparatus to collect wrist vibration data from 20+ participant trials
- Laboratory experience through collecting bioplastic material data, published research in **Composites Part C**
- Proficient in **SOLIDWORKS & 3D printing** for low-profile ankle brace, and completed CSWA Associate Certification

Education

Master of Applied Science (MASc., Computer Engineering) + Specialization in Artificial Intelligence
University of Guelph, in collaboration with Vector Institute, *Sep. 2024 – present*
Thesis Project: Hand-Arm Vibration Identification in the Workplace using Machine Learning Methods
Relevant Coursework: Biomedical Robotics, Introduction to Machine Learning

Bachelor of Engineering (BEng., Biomedical, w/co-op)
University of Guelph, *Sep. 2019 – Apr. 2024*
Relevant Coursework: Biomechanical Engineering Design, Robotic Systems, Medical Image Modalities

Work Experience

Graduate Researcher – Thesis Project - Hand-Arm Vibration in the Workplace using Machine Learning Methods
University of Guelph, Guelph, ON, Canada, *September 2024 – present*

- Identified gaps in assessing hand-arm vibration risk for power tool operators based on existing standard ISO 5349-1
- Developed **statistical (Excel, IBM SPSS) and machine learning (Python)** pipelines for predicting cumulative hand-arm vibration exposure using 6000+ vibration data points
- Designed **electronics-based** accelerometer and force plate testing apparatus following **ISO 5349** to collect wrist vibration data on **20+ participants**
- Presented research at **Ontario Biomechanics Conference (2025)** and accepted to present at **World Congress of Biomechanics (2026)**

Medical Device Repair Technician (Co-Op)
Stryker, Waterdown, ON, Canada, *May 2023 – December 2023*

- Provided **function checks** on Endoscopy, Instruments, and Navigation medical equipment, meeting **99.8%** accuracy requirements, including familiarity with **regulatory standards**
- **Diagnosed, repaired** and **calibrated electromechanical** medical devices **in-house and in field (hospital)** settings (ex. surgical drills, insufflator consoles), adhering to **standard operating procedures**, meeting **repair-time** targets
- Improved QA processes by updating documents and validating testing fixtures for equipment verification workflows
- Generated customer repair documentation and service quotes using **Oracle ERP** platform
- Developed **HTML-based** communication templates within **Salesforce** and **Prontoforms** platforms in partnership with Service Sales team, standardizing customer outreach with corporate branding guidelines

Undergraduate Researcher (Co-Op) – Bioplastic Blends
Bioproducts Discovery & Development Centre (BDDC), University of Guelph, Guelph, ON, January 2022 – April 2022

- **Primary author** of a **published manuscript** in Composites Part C, "Sustainable Biocomposites from Biofuel Co-Product and Biodegradable Plastic: Effect of Pyrolysis and Compatibilizer on Performance" (https://doi.org/10.1016/j.jcomc.2022.100301)
- - Used **FTIR Spectroscopy,** mechanical analysis (**INSTRON, Zwick Impact Tester),** thermogravimetric analysis **(TA Q500 thermal analyzer),** and rheology to determine mechanical and thermal properties
- Designed **bio-plastic composition** to improve sustainability of blister packs for **Project Soy Competition 2023**, **2nd place** undergraduate team

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Projects

Senior Capstone Project – Low Profile Ankle Brace
University of Guelph, Guelph, ON, Canada, *August 2023 – April 2024*

- Designed low-profile ankle brace prototype to fit within tight-fitting footwear for high-performance athletes using **3D printing mesh structures** using **SolidWorks**
- Validated **material properties** and different mesh structures using **INSTRON** tensile testing and designed **3D-printed** simulated foot to test effects on ankle mobility
- Prepared **regulatory filings** to the Research Ethics Board to gain approval for low-impact human testing

Biomechanical Design Project – Improving Brake Pedal Realism in Driving Simulator
University of Guelph, Guelph, ON, Canada, *January 2024 – April 2024*

- Investigated **ergonomics** of car brake pedals using the University of Guelph DRIVELAB driving simulator to design a brake without use of hydraulic fluids due to simulator instrumentation restrictions
- Used **VICON** and **EMG** sensors to determine applied force and muscle activation on **fully mechanical** brake

Bioinstrumentation Design – Water Safety Detection Cap
University of Guelph, Guelph, ON, Canada, *October 2022 – December 2022*

- Collaborated in **team of 4** to modify swim cap with **Arduino** compatible sensors and **Bluetooth** module to alert lifesaving personnel in drowning situation
- Received 3rd place prize out of 17 groups on Guelph Engineering Design Day for innovation and presentation

Leadership

Graduate Teaching Assistant (GTA)
Biomechanical Engineering Design *(W26),* Biomaterials *(W25)*, Engineering Design I Team Lead *(F24/F25)*
University of Guelph, Guelph, ON, Canada, *September 2024 – April 2026*

- Supported **800+ students** through teaching introductory circuits lab and **troubleshooting** design projects in weekly office hours, emphasizing design process and collaborative approach
- Supported **final-year** client-centered design projects involving **CAD (SolidWorks), biomechanical analysis (MATLAB)** and **data collection (VICON Nexus, IMU),** focusing on key engineering **design principles**
- Team Lead TA for Engineering Design I, coordinating **team of 4** undergraduate TAs

Guelph Engineering Society
University of Guelph, Guelph, ON, Canada, *September 2022 – April 2024*

Special Events Coordinator

- Organized **Orientation Week** events for incoming first-year engineering students

Thornborough Manager

- Managed **Engineering Equipment Library**, an initiative for engineering students to reuse design project materials
- Organized internal Engineering society functions for 30+ members, including locker distribution, office hours, and communicating student body concerns to the building manager

Children's Program Volunteer – Quadball
City of Guelph, Guelph, ON, Canada, *September 2022 – March 2024*

- Led weekly children's session teaching them about Quadball in a fun and energetic setting

Digital Development Coordinator
Tootyr (NewArts), Toronto, ON, Canada, *July 2020 – August 2022*

- **Managed volunteer teams** at **charity start-up** working in fundraising & event planning**, hosting weekly meetings**
- Wrote **fundraising grants** for start-up education charity during Covid-19 targeted at **corporate foundations**
- Redesigned the Tootyr website using **HTML & Javascript** to improve the user experience for both students and tutors, implementing **automated tutor onboarding** and adding **subscription functionality**

Interests

- Human-centred design in the medical devices space, such as focusing on **reducing injury risk**
- Integrating **medical robotics and machine learning knowledge** in the medical devices space
- Community involvement, specifically in **engineering education** and **promoting healthy lifestyles**
- Sports, including playing Quadball for Team Canada, as well as ultimate frisbee, hockey and soccer